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Google Scholar link:

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PROFESSIONAL PROFILE

A result-oriented Research Scientist with expertise in mechanical properties of materials (statistical modelling and experiments) and a keen interest in developing sustainable and eco-friendly building materials targeted at solving societal infrastructural challenges in a sustainable manner.

Key Competencies include ability to develop and deliver learning/teaching materials, strong data analytical skills, good technical reading skills, ability to manage project costs and quality successfully, good networking skills, ability to produce original research for publication, ability to build and lead an independent research program, high level of integrity and ability to work with minimum supervision, good problem-solving skills and initiative, ability to manage short and long-term deadlines and projects, proficient in the use of Microsoft office tools and AutoCAD, experience in report/proposal writing, presentation writing and delivery.

ACADEMIC HISTORY

African University of Science and Technology, Abuja

- 2020 (Ph.D. in Materials Science and Engineering – Sustainable Building Materials Option)

African University of Science and Technology, Abuja

- 2017 (M.Sc. in Materials Science and Engineering – Sustainable Building Materials Option)

University of Nigeria Nsukka, Enugu State

- 2010 (B.Eng. in Civil Engineering)

National Open University of Nigeria, Abuja

- 2015 (Post Graduate Diploma in Education)

University of Nigeria Nsukka, Enugu State

- 2006 (National Diploma in Science Laboratory Technology)

WORK EXPERIENCE

Postdoctoral Researcher
2022)

(January 2021 – May

Department of Materials Science and Engineering, African University of Science and Technology, Abuja.

- Conducted high-quality research on solving national and global infrastructural challenges.

- Taught Mechanical Behaviour of Materials Course to graduate students.
- Performed independent and collaborative experiments resulting in peer-reviewed publications.
- Prepared and graded examinations, laboratory assignments and reports.
- Participated in Conference and Seminar presentations and assisted in coordinating MSc projects.
- Counselling students on course and career decisions.
- Delivered professional consultative services to government, industry and private individuals.

Research Assistant
September 2020)

(October 2019 –

Department of Materials Science and Engineering, African University of Science and Technology, Abuja.

- Prepared presentations for seminars, conferences, symposiums and workshops.
- Developed laboratory manuals and student report templates.
- Assisted in the teaching of M.Sc. courses and coordination of MSc research projects
- Assisted in women in STEM research and developmental projects
- Competed and won RILEM PhD Travel Grant 2020 (International Conference, Sheffield University, UK.)
- Successful development and submission of Research/Project Proposals

Intern/ Support Staff

(April 2019 – September 2019)

Nigerian National Petroleum Corporation (NNPC), NNPC Towers, Herbert Macaulay Way, Central Business District, Abuja.

- Executed materials procurement procedures productively;
- Managed Vendors registration documents and database successfully;
- Conducted Contract Bidding processes such as Bids Advert and Bids Opening;
- Conducted Stock Replenishments and Stock write-off successfully.

Quality Control Supervisor
2012 – 2015)

(August

BUILDFORT CONSTRUCTIONS LTD , 23, Waziri Ibrahim Crescent, Off Abdulsalami Abubakar Way, Gudu District, Abuja.

- Successful development of Fort Royal Home Estate, Airport Road, Abuja.
- Directed workers engaged in tabulating data, measuring and testing products successfully.
- Managed quality control engineering, such as design and product evaluation.
- Supervised block works and steel reinforcements.

Mathematics Teacher (Teaching Practice) (September – December 2014)

JUNIOR SECONDARY SCHOOL, GOSA, Airport Road, Abuja.

- Developed lesson plans for teaching Mathematics and Instructed students on the subject of Mathematics
- Successfully managed students in the classroom and prepares them for a standardized test.
- Inculcated arithmetic and problem-solving skills in the students successfully.

Assistant Project Supervisor (NYSC) (August 2011 – July 2012)

BUILDFORT CONSTRUCTIONS LTD, 23, Waziri Ibrahim Crescent, Off Abdulsalami Abubakar Way, Gudu District, Abuja.

- Constructed 3- and 2-bedroom bungalows in street blocks, Medical Centre, Shopping mall and Administrative complex.
- Supervised concrete casting operations (Slabs, Beam and Column) of buildings.
- Successfully produced setting out and distribution of levels for drains, gully pot and chambers.
- Supervised earthmoving equipment and road construction successfully.

Estate & Works Personnel (SIWES) (Jan– Dec. 2009)

FEDERAL MINISTRY OF SCIENCE & TECHNOLOGY (Raw Materials Research & Development Council), Abuja.

- General maintenance of infrastructural facilities
- Coordinated work with other disciplines such as architectural, mechanical, etc.
- Handled administrative duties.

PROFESSIONAL QUALIFICATION/TRAINING

- Programming with Python (**Public Policy in Africa Initiative (PPIAI Coding Training)**)-2021
- Research Writing in Sciences (**AuthorAID project of INASP, UK**) – 2019
- Certificate in Health Safety and Environment (HSE) (**Dexter & Heroes Consulting Ltd, Abuja**) - 2012
- Executive Master's Certificate in Project Management using **Prince2** methodology and Certificate in Microsoft Project 2007 (**Pneuma Project Academy: A subsidiary of Pneuma Project Experts Limited**) - 2011

PUBLICATIONS

- 1) **Obianyo, I.I.**; Mahamat, A.A.; Anosike-Francis, E. N.; Stanislas, T.T.; Onyelowe, K.C.; Onwualu, A.P.; Soboyejo, A.B.O. Performance of lateritic soil stabilized with combination of bone and palm bunch ash for sustainable building applications. *Cogent Engineering*. **2021**, 8 (1) 1921673. <https://doi.org/10.1080/23311916.2021.1921673>.
- 2) **Obianyo, I. I.**; Anosike-Francis, E. N.; Odochi, G.; Geng, Y.; Jin, R.; Peter, A.; Soboyejo, A. B. O. Multivariate Regression Models for Predicting the Compressive Strength of Bone Ash Stabilized Lateritic Soil for Sustainable Building. *Construction and Building Materials*. **2020**, 263, 120677.

<https://doi.org/10.1016/j.conbuildmat.2020.120677>. Elsevier Science Publishers, Amsterdam, the Netherlands.

- 3) **Obianyo, I. I.**; Onwualu, A. P.; Soboyejo, A. B. O. Mechanical Behaviour of Lateritic Soil Stabilized with Bone Ash and Hydrated Lime for Sustainable Building Applications. *Case Studies in Construction Materials*. **2020**, *12*, e00331. <https://doi.org/10.1016/j.cscm.2020.e00331>. Elsevier Science Publishers, Amsterdam, the Netherlands.
- 4) **Obianyo, I. I.**; Ihekwe, G. O.; Mahamat, A. A.; Onyelowe, K. C.; Onwualu, A. P.; Soboyejo, A. B. O. Overcoming the Obstacles to Sustainable Housing and Urban Development in Nigeria: The Role of Research and Innovation. *Clean. Eng. Technol.* **2021**, *4* (December), 100226. <https://doi.org/10.1016/j.clet.2021.100226>.
- 5) **Obianyo, I. I.**; Mahamat, A.A.; Stanislas, T. T.; Ihekwe, G. O.; Kelechi, S.E.; Onyelowe, K. C.; Onwualu, A. P.; Soboyejo, A. B. O. Production and Utilization of Earth - Based Bricks for Sustainable Building Applications in Nigeria: Status, Benefits, Challenges and Way Forward. *J. Build. Pathol. Rehabil.* **2021**, *4*, 1–11. <https://doi.org/10.1007/s41024-021-00131-4>.
- 6) Adedeji, L.; **Obianyo, I. I.**; Olalekan, A.; Peter, A.; Soboyejo, A. B. O.; Amu, O. O. Case Studies in Construction Materials Mechanical Behaviour of Composite Produced with Quarry Dust and Rice Husk Ash for Sustainable Building Applications. *Case Stud. Constr. Mater.* **2022**, *17* (April), e01157. <https://doi.org/10.1016/j.cscm.2022.e01157>.
- 7) Anosike-francis, E. N.; **Obianyo, I. I.**; Wasiu, O.; Odochi, G.; Ikpi, M.; Olajide, A.; Peter, A. Physical-Mechanical Properties of Wood Based Composite Reinforced with Recycled Polypropylene and Cowpea (*Vigna Unguiculata* Walp .) Husk. *Clean. Mater.* **2022**, *5* (May), 100101. <https://doi.org/10.1016/j.clema.2022.100101>.
- 8) Kelechi, S. E.; Adamu, M.; Uche, O. A. U.; Okokpujie, I. P.; Ibrahim, Y. E.; **Obianyo, I. I.** A Comprehensive Review on Coal Fly Ash and Its Application in the Construction Industry. *Cogent Eng.*, **2022**, *9* (1), 0–26. <https://doi.org/10.1080/23311916.2022.2114201>.
- 9) Al Biajawi, M. I.; Embong, R.; Muthusamy, K.; Ismail, N.; **Obianyo, I. I.** Recycled Coal Bottom Ash as Sustainable Materials for Cement Replacement in Cementitious Composites: A Review. *Constr. Build. Mater.* **2022**, *338*, 127624. <https://doi.org/10.1016/j.conbuildmat.2022.127624>.
- 10) Aboubakar, A.; Dayyabu, A.; Sanusi, A.; Ado, M.; **Obianyo, I. I.**; Tiwa, T.; Linda, N. Heliyon Dimensional Stability and Strength Appraisal of Termite Hill Soil Stabilisation Using Hybrid Bio-Waste and Cement for Eco-Friendly Housing. *Heliyon*, **2022**, *8* (May), e09406. <https://doi.org/10.1016/j.heliyon.2022.e09406>.
- 11) Kelechi, S. E.; Uche, O. A. U.; Adamu, M.; Alanazi, H.; Okokpujie, I. P.; Ibrahim, Y. E.; **Obianyo, I. I.** Modeling and Optimization of High-Volume Fly Ash Self-Compacting Response Surface Methodology. *Arab. J. Sci. Eng.* **2022**, May. <https://doi.org/10.1007/s13369-022-06850-1>.

- 12) Kelechi, S. E.; Adamu, M.; Mohammed, A.; **Obianyo, I. I.**; Ibrahim, Y. E. Equivalent CO₂ Emission and Cost Analysis of Green Self-Compacting Rubberized Concrete. *Sustainability*, **2022**, *14* (137), 1–14. <https://doi.org/https://doi.org/10.3390/su14010137>.
- 13) Anosike-francis, E. N.; Ubi, P. A.; **Obianyo, I. I.**; Kalu-uka, G. M.; Bello, A.; Ofem, M. I.; Olorunnisola, A. O.; Onwualu, A. P. Mechanical and Thermomechanical Properties of Clay-Cowpea (*Vigna Unguiculata* Walp) Husks Polyester Bio-Composite for Building Applications. *Appl. Sci.*, **2022**, *12* (713), 1–17. <https://doi.org/https://doi.org/10.3390/app12020713>.
- 14) Kelechi, S. E.; Adamu, M.; Ibrahim, Y. E.; **Obianyo, I. I.** Durability Performance of Self-Compacting Concrete Containing Crumb Rubber, Fly Ash and Calcium Carbide Waste. *Materials (Basel)*, **2022**, *15* (488). <https://doi.org/https://doi.org/10.3390/ma15020488>.
- 15) Onyelowe, K. C., & **Obianyo, I. I.** Influence of moisture and geofluids (GF) on the morphology of quarry fines treated lateritic soil. *Cleaner Engineering and Technology*. **2021**, *3*, 100120. Elsevier Science Publishers, Amsterdam, the Netherlands. <https://doi.org/10.1016/j.clet.2021.100120>.
- 16) Mahamat, A. A., Boukar, M. M., Ibrahim, N. M., Stanislas, T. T., Bih, N. L., **Obianyo, I. I.**, & Savastano, H. Machine Learning Approaches for Prediction of the Compressive Strength of Alkali Activated Termite Mound Soil. *Applied Sciences*. **2021**, *11*(4754). <https://doi.org/10.3390/app11114754>.
- 17) Ihekweme, G. O., **Obianyo, I. I.**, Orisekeh, K. I., Kalu-uka, M., & Nwuzor, I. C. Plasticity characterization of certain Nigeria clay minerals for their application in ceramic water filters. *Science Progress*. **2021**, *104*(2), 1–13. <https://doi.org/10.1177/00368504211012148>.
- 18) Ihekweme, G. O.; **Obianyo, I. I.**; Anyakora, N.; Odusanya, O. S.; Onwualu, A. P. Expanded Clay Aggregates Multi-Functionality for Water Purification : Disinfection and Adsorption Studies. *Cogent Engineering*. **2021**, *8* (1). <https://doi.org/10.1080/23311916.2021.1883232>.
- 19) Onyelowe, K. C.; Onyia, M. E.; Nguyen-thi, D.; Van, D. B.; Onukwugha, E.; Baykara, H.; **Obianyo, I. I.**; Dao-phuc, L. Swelling Potential of Clayey Soil Modified with Rice Husk Ash Activated by Calcination for Pavement Underlay by Plasticity Index Method (PIM). *Advances in Materials Science and Engineering*. **2021**, *2021* (January). <https://doi.org/10.1155/2021/6688519>.
- 20) Aboubakar, A.; **Obianyo, I. I.**; Ngayakamo, B.; Linda, N.; Ayeni, O.; Azeko, S. T.; Savastano, H. Alkali Activation of Compacted Termite Mound Soil for Eco-Friendly Construction Materials. *Heliyon*. **2021**, *7* (March), e06597. <https://doi.org/10.1016/j.heliyon.2021.e06597>.
- 21) Onyelowe, K. C.; Onyelowe, F.A.D.; Van, D. B.; **Obianyo, I. I.**; Kelechi, E. Overview of Ash as Supplementary Cementitious Silicate-Based Composite and Construction Material. *Építőanyag Journal of Silicate Based and Composite Materials*. **2020**, *72* (3).
- 22) Stanislas, T.T.; Komadja, G.C.; Ngasoh, O.F.; **Obianyo, I.I.**; Tendo, J.F.; Onwualu, A.P.; Savastano, H. Physical-mechanical Performance and

Durability of Cellulose Pulp Reinforced Extruded Earth-based Composites. *Arabian Journal for Science and Engineering*. **2021**, 0123456789. <https://doi.org/10.1007/s13369-021-05698-1>.

- 23) Onyelowe, K.; Onyia, M.; Onukwugha, E.; Ezugwu, C.; Odumade, A.; **Obianyo, I.I.**; Chigbo, I.; Kelechi-Asumba, S. Atterberg Limits of Modified Compacted Clayey Soil for Sustainable Green Subgrade Structure. **2021**, 33 (3). <https://doi.org/10.17576/jkukm>.
- 24) Onyelowe, K. C.; **Obianyo, I. I.**; Onwualu, A. P.; Onyia, M. E.; Moses, C. Morphology and Mineralogy of Rice Husk Ash Treated Soil for Green and Sustainable Landfill Liner Construction Morphology and Mineralogy of Rice Husk Ash Treated Soil for Green and Sustainable Landfill Liner Construction. *Clean. Mater.* **2021**, 1 (December), 100007. <https://doi.org/10.1016/j.clema.2021.100007>.
- 25) Onyelowe, K. C.; Ebid, A. M.; Nwobia, L. I.; **Obianyo, I. I.** Shrinkage Limit Multi - AI - Based Predictive Models for Sustainable Utilization of Activated Rice Husk Ash for Treating Expansive Pavement Subgrade. *Transp. Infrastruct. Geotechnol.* **2021**, No. 0123456789. <https://doi.org/10.1007/s40515-021-00199-y>.

GRANTS AND FELLOWSHIP

1. PAMI Scholarship for Master's Degree (2016);
2. Research Grant for M.Sc. Thesis (2017);
3. PAMI Scholarship for PhD (2018);
4. Travel Grant (2017) for AMRS International Conference held at Botswana;
5. Travel Grant (2019) for AMRS International Conference held at Tanzania;
6. PAMI Internship grant (2019);
7. RILEM PhD Grant (2020) for RILEM International Conference held at Sheffield University, UK;
8. PAMI Visiting Scholar Fellowship (2020) to Ohio State University, USA.

MEMBERSHIP OF LEARNED SOCIETIES AND PROFESSIONAL BODIES

- RILEM (The International Union of Lab. Experts in Construction Materials, Systems & Structures)
 - OWSD (Organization for Women in Science for the Developing World)
 - MRS (Materials Research Society);
 - AMRS (African Materials Research Society)
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VOLUNTEERING AND MENTORSHIP

- Volunteer at Girls in STEM Outreach, African University of Science and Technology: 2021
- Volunteer at African Youth and Talent Summit (AYTS) and Team Breed, PCI Nigeria: 2020 - till date
- Volunteer at African University of Science and Tech. for Secondary School Science Competition: 2020

- Volunteer at NIMACON International Conference, 2019; Volunteer at KC Mentorship Platform (KMP): 2020
 - Volunteer at AuthorAID (Project of INASP, UK) Mentorship Platform: 2019 – till date
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REFEREES

Prof. Azikiwe Peter Onwualu aonwualu@aust.edu.ng; +2348037432497
Acting President, African University of Science and Technology, Abuja, Nigeria.

Dr. Mofoluwake Ishola mofoluwake.ishola@scanacon.com; +46769788617
Senior Research and Development Engineer, Scanacon, Stockholm Metropolitan Area, Sweden.

Dr. Bello Abdulhakeem abello@aust.edu.ng; +2348038227233
Resident Faculty, Theoretical and Applied Physics, African University of Science and Technology, Abuja, Nigeria.